M.TECH (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING). GROUP - 13

Mulberry Leaf Classification

PRESENTED BY : TRIPTI MISHRA (MT/AI/10032/26)

RISHIKA RAJ (MT/AI/10010/26)



REAL-WORLD PROBLEM

Real-world Problem

Similar leaf appearance makes manual variety identification difficult. We test image-based classification on actual mulberry leaf data.

Research question: Does Curriculum Learning beat Random Training?

HYPOTHESIS 01

ImageNet-pretrained transfer learning can learn useful visual features.

HYPOTHESIS 02

Easier-to-harder ordering may improve performance.

- Closely related mulberry varieties can look very similar to the eye.
- Identifying one of 10 varieties from leaf appearance alone is difficult.
- We test whether a CNN can learn useful visual distinctions from real leaf images.



One leaf image carries the variety signal.

RESEARCH QUESTION

Research Question & Hypothesis

Can EfficientNetB0 classify 10 mulberry leaf varieties, and does Curriculum Learning outperform Random Training?

H1

ImageNet-pretrained EfficientNetB0 can learn useful visual representations.

H2

Training on easier examples before harder examples may improve performance.

H3

Random training may provide more consistent classification performance across different random seeds than curriculum learning.

Dataset Description

5,262 total images

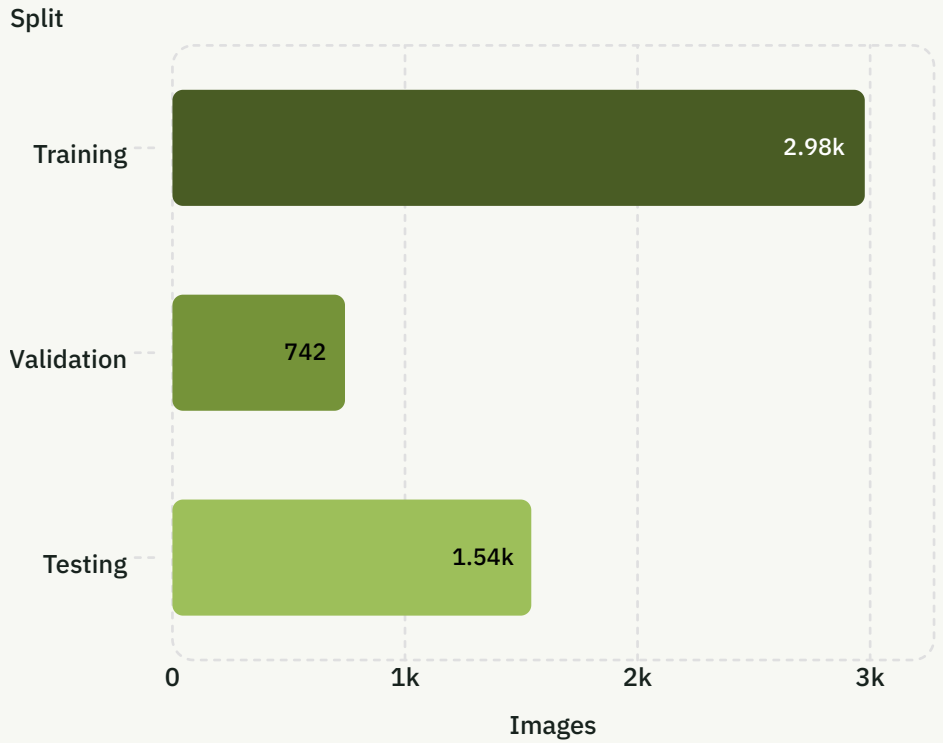
2,977 training · 742 validation · 1,543 testing

Inputs

224 × 224 RGB

10 varieties

Uneven test-set class counts; class-wise evaluation matters.



Classes: ChiangMai60 · RedKing · WhiteKing · BlackOodTurkey · TaiwanStraberry · BlackAustralia · Buriram60 · Kamphaengsaeng42 · TaiwanMeacho · ChiangMaiBuriram60

Methodology

Baseline workflow

1 · Input

Input Leaf Image

2 · Preprocessing

Resize to 224 × 224 × 3

3 · EfficientNetB0

ImageNet-pretrained backbone, **frozen**

4 · Feature Extraction or Classification Head

GAP → Dense 128 ReLU → Dropout 0.3 → Dense 10 Softmax

5 · Variety Prediction

Predicted mulberry variety

Training: Adam · learning rate 0.001 · categorical cross-entropy · batch size 32 · Early Stopping

Parameters: 4,214,829 total · 165,258 trainable

Curriculum process (Experiment B)

Difficulty estimation

Easier samples first

Progressively harder samples

Evaluation

Experimental Setup

EXPERIMENT A

Image-level baseline

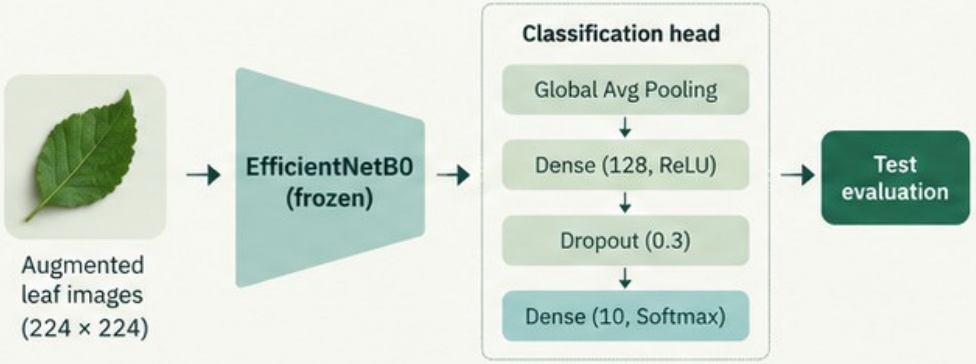
Augmented leaf images → Frozen EfficientNetB0 → Trained classification head → Validation monitoring and held-out test

EXPERIMENT B

Feature-extraction comparison

1,280-dimensional EfficientNetB0 features → Centroid cosine-similarity difficulty → Random Training · Curriculum Learning · Curriculum-Shuffled

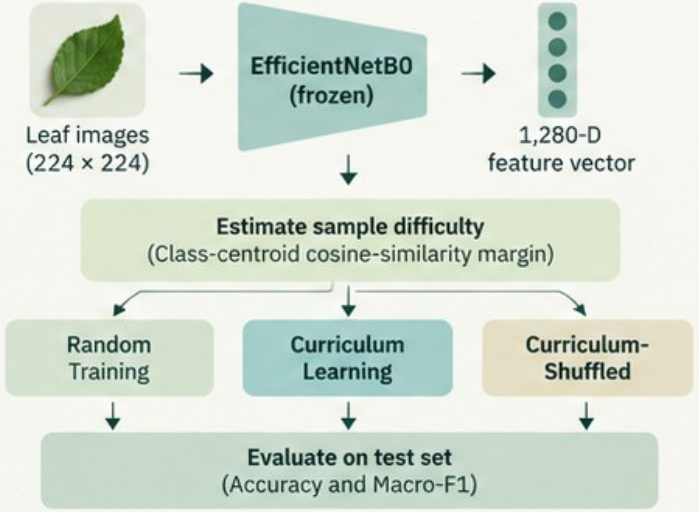
EXPERIMENT A · IMAGE-LEVEL BASELINE PIPELINE



Key settings (Experiment A)

- Pretrained weights: ImageNet
- Backbone: Frozen
- Optimizer: Adam (lr = 0.001)
- Loss function: Categorical cross-entropy

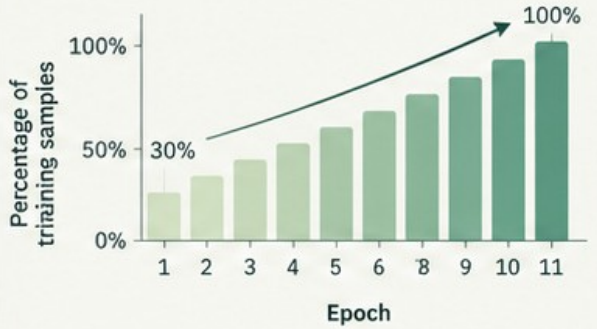
EXPERIMENT B · FEATURE-EXTRACTION PIPELINE



Experimental details (Experiment B)

- Random seeds: 0, 1, and 2 (for each strategy)
- Feature dimension: 1,280
- Difficulty measure: Class-centroid cosine-similarity margin

Curriculum progression schedule



Schematic illustration of curriculum schedule: 30% → 100% by epoch 11 (intermediate percentages not measured).

Key control

The two experiments use different pipelines. Baseline results are not the same measurement as the strategy comparison.

EXPERIMENTAL RESULTS

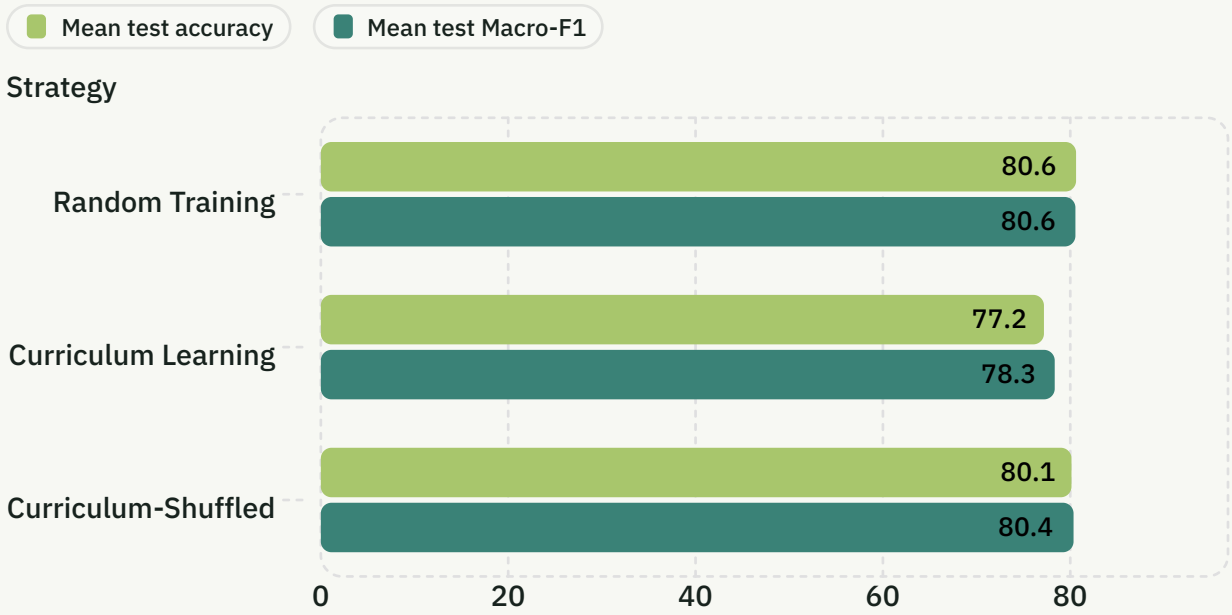
Experimental Results

77.64%
Baseline test accuracy

77.76%
Validation accuracy

77.75%
Validation Macro-F1

76.84%
Test Macro-F1



Random Training achieved the highest mean test accuracy (80.62%)

in the feature-extraction comparison.

Curriculum-Shuffled 80.13% · Curriculum Learning 77.19%

Means across three random seeds. This is a different pipeline from the baseline.

RESULTS

Model Performance — Accuracy & Macro-F1

Baseline EfficientNetB0

Image-level EfficientNetB0 baseline on the held-out test set.

Validation Accuracy

77.76%

Validation Macro-F1

77.75%

Test Accuracy

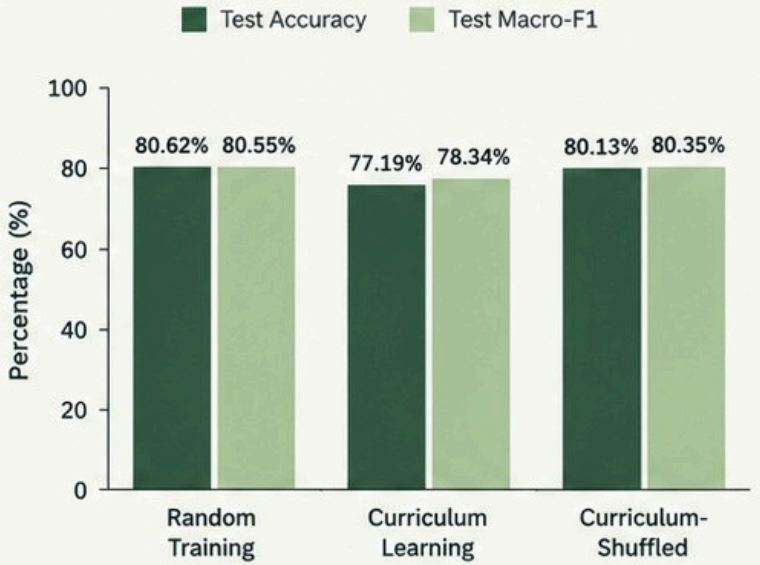
77.64%

Test Macro-F1

76.84%

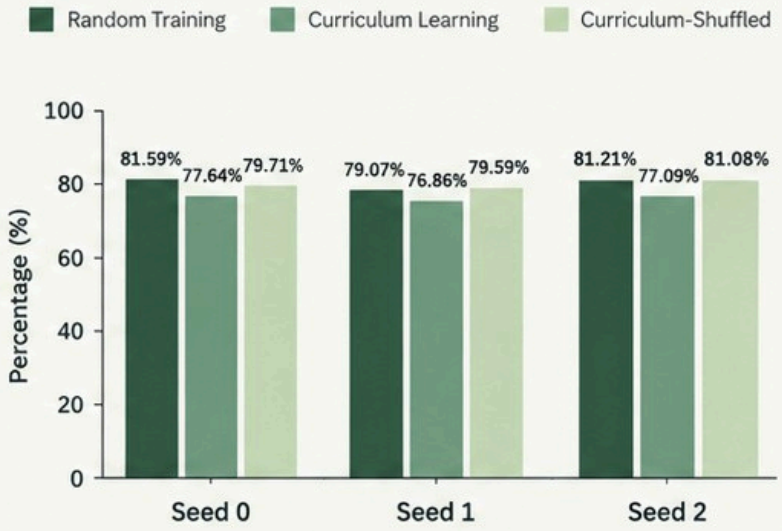
Training Strategy Comparison

Mean test accuracy and Macro-F1 across three random seeds from the feature-extraction experiment.



Test Accuracy Across Seeds

Test accuracy for each training strategy across three random seeds (feature-extraction experiment).

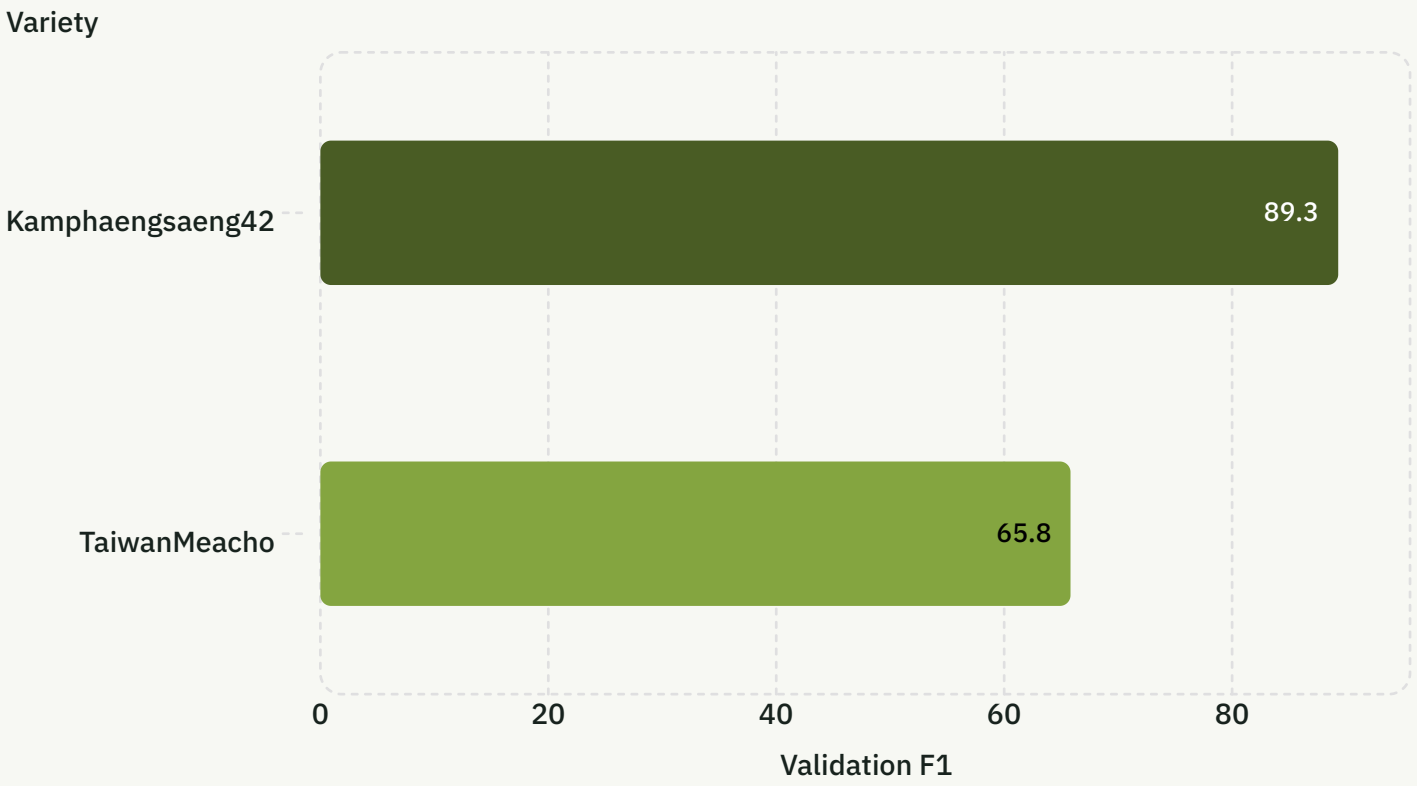


KEY TAKEAWAY

Random Training achieved the highest mean test accuracy and Macro-F1 in the strategy comparison.

⚠ Baseline and strategy-comparison experiments use different pipelines; do not compare them as a controlled head-to-head experiment.

Interesting Findings & Failure Cases



89.31%

Kamphaengsaeng42 validation F1

65.82%

TaiwanMeacho validation F1

- Performance varies across varieties.
- Curriculum Learning did not outperform Random Training.
- The difficulty estimate is heuristic-based.

CONCLUSION

Conclusion

77.64%

Baseline test accuracy

Image-level EfficientNetB0 baseline on the held-out test set.

- EfficientNetB0 with ImageNet-pretrained weights.
- Frozen backbone with a trained classification head.
- Test Macro-F1: **76.84%**.

Takeaway: The baseline learned useful visual features for classifying 10 mulberry leaf varieties.

80.62%

Random Training mean test accuracy

From the separate feature-extraction experiment.

- Mean test Macro-F1: **80.55%**.
- Evaluated across three random seeds: 0, 1, and 2.
- Curriculum-Shuffled achieved 80.13% mean test accuracy.

Takeaway: Random Training achieved the best mean test performance among the three strategies evaluated.

Not better

Curriculum Learning did not outperform Random Training

Under the tested setup, its mean test accuracy was 77.19%.

- Mean test accuracy: **77.19%**.
- Mean test Macro-F1: **78.34%**.
- Easier-to-harder ordering did not improve performance under the tested setup.

Takeaway: Curriculum learning does not guarantee better classification performance.

Limitations & Future Improvement

Limitations

1. **Class Imbalance**
   Uneven class distribution may affect classification performance across mulberry varieties.
2. **Heuristic Difficulty Estimation**
   Sample difficulty is estimated using a class-centroid cosine-similarity margin.
3. **Limited Random Seeds**
   The training strategy comparison uses only three random seeds: 0, 1, and 2.
4. **Different Experimental Pipelines**
   The baseline and training strategy comparison use different pipelines, limiting direct comparison.
5. **Limited Generalization Testing**
   Performance has not been evaluated on an independent external dataset.

→

Current limitation → Proposed improvement

Future Improvements

1. **Explore Curriculum Schedules**
   Compare alternative difficulty measures and easier-to-harder sample schedules.
2. **Increase Experimental Reliability**
   Run more experiments using additional random seeds.
3. **Fine-Tune EfficientNetB0**
   Investigate fine-tuning the pretrained backbone as a separate experiment.

NEXT STEP

Thank You | Questions & Discussion